Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/04/2005
PI Name:	Green, Jason Ph.D.		
Project Title:	Fundamentals of Ion- Molecule/Surface Reactions and Instrument Development		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2084	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	By advancing mass spectrometric instrumentation and understanding of fundamental aspects of gas-phase ion/molecule and ion/surface reactions we contribute to NASA's effort to characterize astrophysical and atmospheric environments. Progress has been made in ongoing research of (1) an anomalous heavy-atom kinetic isotope effect and (2) the development of a new instrument amenable to production of isotopically enriched species by this isotope effect and collection of this material by ion/surface collisions. These endeavors represent our effort to push the boundaries of fundamental physical science on this planet and beyond.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Our study of kinetic isotope effects of chloroalkanes is of relevance to earth and atmospheric science. Chlorinates solvents degrade ozone in amounts second only to CFCs.
Task Progress:	Advancement of mass spectrometric instrumentation and understanding of fundamental aspects of gas-phase ion-molecule and ion-surface reactions has been and remains our main objective. Our goals primarily contribute to NASA's effort to characterize astrophysical and atmospheric environments. Progress has been made in ongoing research of a newly discovered anomalous heavy-atom kinetic isotope effect in the unimolecular dissociation of chloroalkane chloride complexes. Unexpected isotope abundances in the mass spectra of these species prompted much work on this striking phenomenon partly due to its unprecedented magnitude but also because of its extreme significance to atmospheric and environmental chemistry. Considerable advancements have been made in the understanding of the mechanism causing this isotope effect through numerous experiments. The effect is so great that it could potentially lead to molecular isotope separation method. Prompted by this potential, a device has recently been constructed with the goal of obtaining proof-of-principle data. We hope to separate isotopic species by molecular methods in this new instrument which will allow subsequent mass spectrometric manipulation and landing on an appropriate surface, ultimately providing useable quantities of isotopically enriched material. The latter step makes use of previous knowledge of and work on ion/surface collisions. Construction of the device being complete, current work focuses on production of isotopic enrichment as well as selection and study of surfaces capable of efficient collection and capture of ionic species.
Bibliography Type:	Description: (Last Updated: 04/13/2017)
Articles in Peer-reviewed Journals	Cooks RG, Jo S-C, Green JR. "Collisions of organic ions at surfaces." Applied Surface Science. 2004 Jun 15;231-232:13-21. http://doi.org/10.1016/j.apsusc.2004.03.017 , Jun-2004
Articles in Peer-reviewed Journals	Green JR, Cooks RG. "Inverse heavy-atom kinetic isotope effects in chloroalkanes." Journal of Physical Chemistry A. 2004 Nov; 08(46), 10039-43. http://dx.doi.org/10.1021/jp046228g , Nov-2004
Articles in Peer-reviewed Journals	Gologan B, Green JR, Alvarez J, Laskin J, Cooks RG. "Ion/surface reactions and ion soft-landing." Physical Chemistry Chemical Physics. 2005;7(7):1490-500. http://dx.doi.org/10.1039/B418056A , Feb-2005
Journal/Magazine covers	Gologan B, Green JR, Alvarez J, Laskin J, Cooks RG. "Cover in journal Physical Chemistry Chemical Physics for article, 'Ion/surface reactions and ion soft-landing.' " Physical Chemistry Chemical Physics. 2005;7(7):1329. http://dx.doi.org/10.1039/B503298A , Feb-2005 , Feb-2005